

Forecasting Inflation for 2026 and Beyond

Finding Sumerized

A Vecm showed that inflation's long run trend is upwards from mid 2025 to 3.5% by 2027 but it fails to anticipate a fuel and housing increase pushing inflation up showing the short run trends cannot be predicted using historical data. Intuition is needed when models become outdated due to unforeseen events and before new data is available, a quantitative method must be devised to predict inflation. Using an IRF and a historical event similar to the current one from 1990, a shock of 19% to the fuel price index would increase inflation by 1.27% percentage points which seems appropriate due to the magnitude of the fuel crisis we are experiencing.

Introduction

Having accurate and reliable inflation forecasts is crucial for all policy makers, businesses and investors. Due to the recent spikes in inflation and the RBA's subsequent spikes in cash rate, it's important to understand what factors influence inflation. Understanding inflation can help predict when the RBA might lower cash rate and stimulate the economy.

Bad forecasting can have immense implications on decision makers so it's important to estimate uncertainty and address limitations where present as to not give inaccurate results.

Method

VECM's are the model of choice and reduce misspecification compared to ARIMA and AR models. VECM's can better understand the relationship of cointegrated variables such as cash rate and inflation and better predict how one variable affects the other in long run trends.

But first we use a VAR model to estimate the number of lags of autoregressors to use.

We start by defining the vector as using the following variables which have at least some long-term relationships with inflation.

$$x = \{ \text{cpi_inflation}, \text{unemployment_rate}, \text{cash_rate} \}$$

Using data starting in 1993Q4 and starting the prediction from 2025Q2. Using VAR_est() in R and taking the top 2 ranking of AIC and BIC to create 4 adequate models with the lower the ranking, the better the fit.

P (Number of Autoregressors)	AIC	BIC
9	339.5688	571.5914
10	341.6142	597.6981
2	391.9916	451.2175
1	432.0430	465.9827

Moving forward, models with 9,10 and 2 lags will be used as they have the best balance between best fit and simplicity with 1 lag being too simple and having too much error due to a much larger AIC.

Checking residuals

A Breuch-Godfrey LM test is used to test if there is serial correlation between lags(Evidence that previous values affect future values). In this test we need p to be greater than 0.05 to reject the null hypothesis that there is no evidence of serial correlation at a 95% confidence interval.

P (Number of Autoregressors)	P-value
2	0.3783
9	0.1066
10	0.3291

All models passed the test comfortably and will move on to testing the Eigenvalue.

Eigenvalue

The purpose of testing the Eigenvalue is to see if the values produce a rank 1 matrix so they are suitable for a VECM's. We also need to ensure they are under 1 so we can forecast accurately and the results do scale towards infinity.

P	Max Eigenvalue
2	0.962828

9	0.974629
10	0.966529

Johansen trace procedure

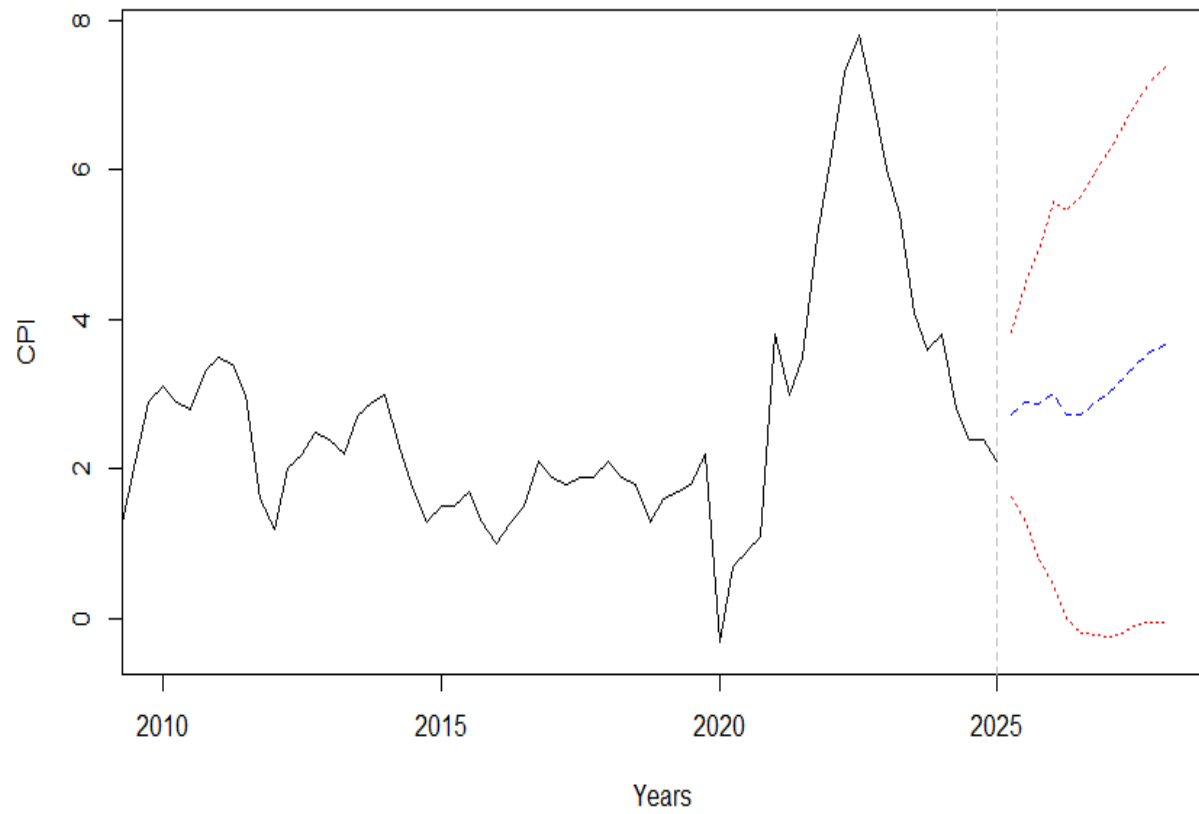
All 3 models show compelling evidence of cointegration at rank 1 and implies a long term relationship between all the variables.

P (Number of Autoregressors)	Test value
2	39.74
9	41.17
10	43.75

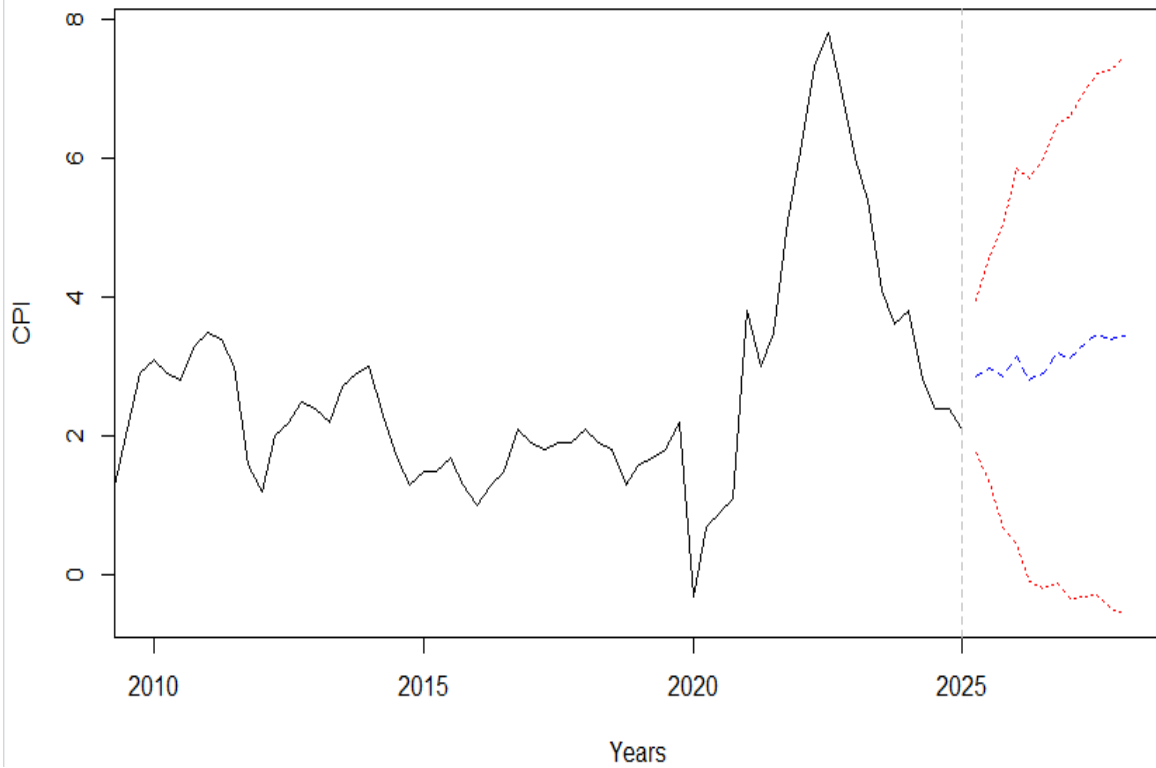
Models 9 and 10 reject no cointegration at the 1% so we will move forward with these 2 models going forward.

Forecasting

Forecast for CPI - VECM, Autolags: 9



Forecast for CPI - VECM, Autolags: 10



Time	9	10
Sep-25	2.736456	2.855002
Dec-25	2.907528	2.974700
Mar-26	2.871281	2.861574
Jun-26	3.018603	3.149385
Sep-26	2.734059	2.805166
Dec-26	2.733944	2.909303
Mar-27	2.889406	3.190519
Jun-27	3.004701	3.136683
Sep-27	3.203347	3.327147
Dec-27	3.408579	3.458430

Mar-28	3.567813	3.389640
Jun-28	3.666022	3.449018

Predictions

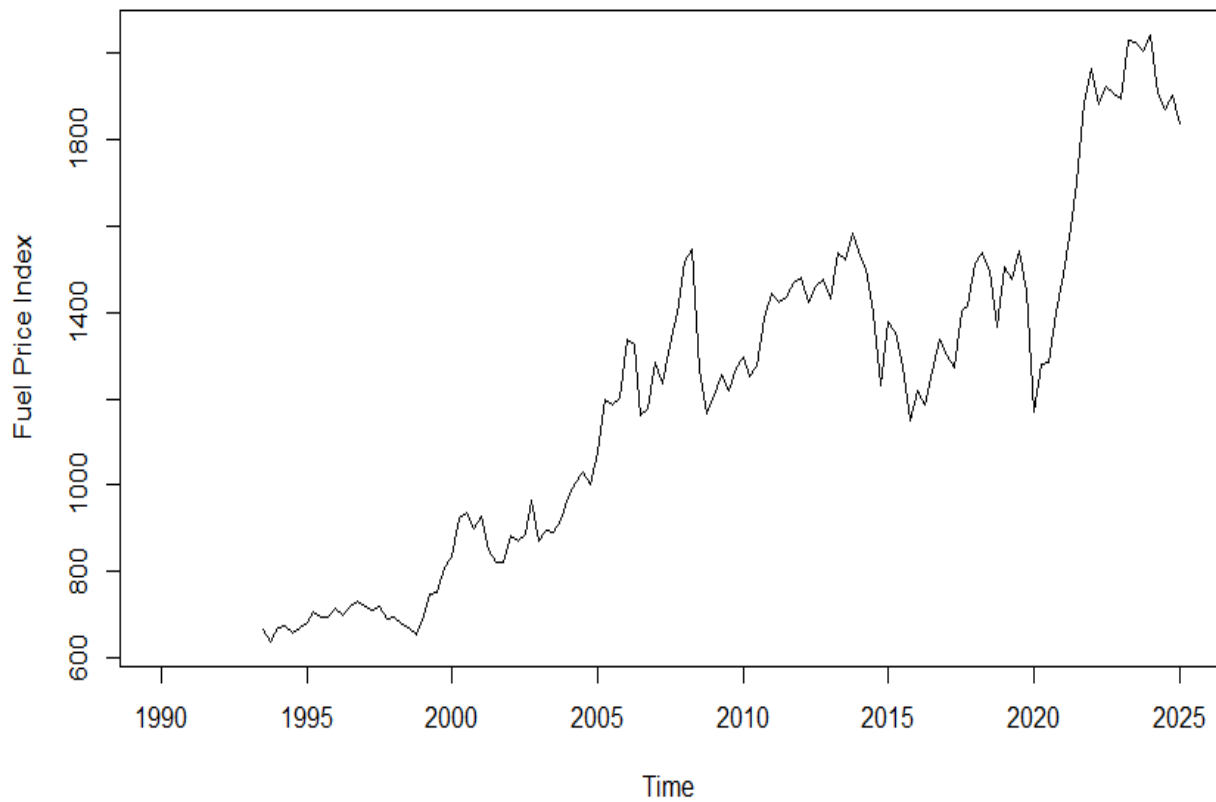
Both models predict an upward trend to around 3.5% despite the mean interest rate and cash rate being lower showing that the VECM takes into account short term trends. Both have wide 95% confidence intervals represented in red lines so reliable forecasts can only be given for 1 to 2 periods. Both models underestimate inflation in the second half of 2025 with inflation rising to 3.6% in September and 3.8% in December driven by a volatile travel and fuel pricing as well as housing proving it is difficult to forecast inflation in the short term due to the many unpredictable factors affecting inflation.

Another factor affecting inflation is the **War in the middle east** affecting fuel prices which is known to have a massive effect on inflation. Unfortunately we can't account for this in the current model but we can account for this in our models. We can also use an impulse response function(IRF) to see how a shock in fuel prices affects inflation for more up to date models.

New model with Fuel prices

Unfortunately we can not use a VECM as the results show that co-integration with the added fuel price index is weak. While the other variables follow long run trends and are affected by each other, fuel price index always increases year on year never reaching a steady equilibrium.

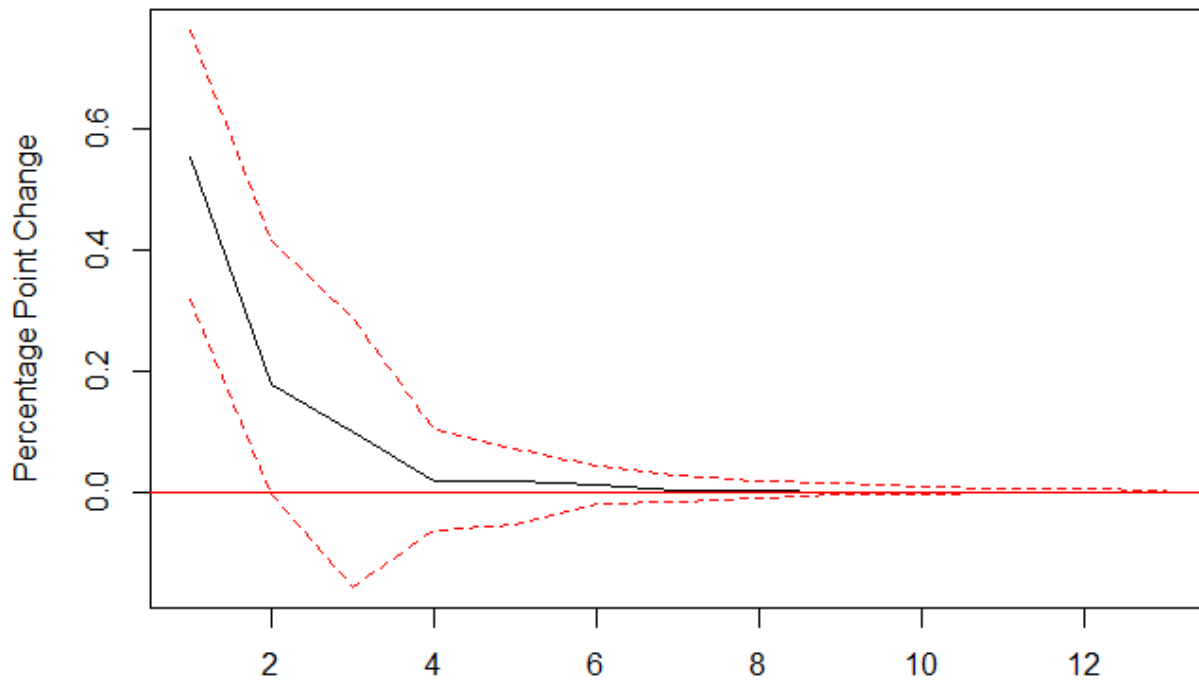
Fuel Prices Over Time



Due to the nature of fuel price index, a Vector Autoregression mode(VAR)l can be used to show how a shock in fuel prices will affect inflation in the short run.

Using an IRF, with a shock of 8.6% in fuel prices from Dec-25 to Mar-26 don't show all impacts of the current fuel crisis with prices peaking in late March and April.

8.6% Shock IRF

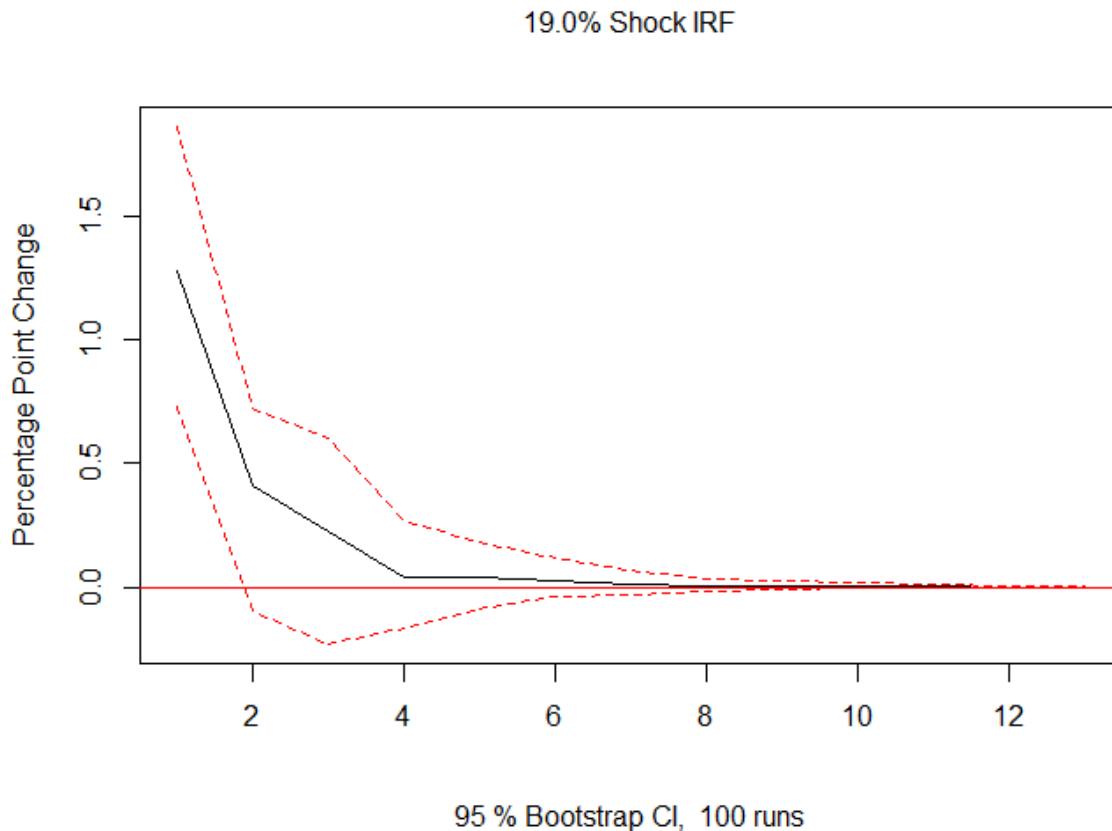


95 % Bootstrap CI, 100 runs

The IRF predicts a 0.55% Percentage point change after the first quarter after the shock so if inflation was 3% before the shock, it would be 3.55% before decaying to the steady state equilibrium after about 4 quarters or a year.

Inflation rose 1.1% from Dec-25 and Mar-26 so the fuel prices can't explain all of the inflation in this time period like housing and other factors but a 0.55% seems low considering how fuel prices almost doubled for a couple months considering how important it is to inflation.

The last time an event affected fuel prices this suddenly was in 1990 what again a war broke out in the middle east but was quickly brought to a close and only affected fuel prices for one quarter. This time the increase in the fuel price was 19% in the fuel index from the last period, while this thinking is somewhat flawed as it doesn't account for how modern Australia functions but is a good historical indicator for what might happen.



This scenario predicts a 1.27% percentage point change and seems more accurate due to the magnitude of the fuel crisis.

Core Findings

After conducting all the necessary tests to prove this data was right to use in a VECM and there was proof of serial correlation and therefore a long run equilibrium. Both adequate models of 9 and 10 lags mostly predicted a gradual upward trend inflation with models peaking at 3.5% by the first half of 2028 which still may be the case but recent events have pushed inflation well above those levels.

Before waiting for more data to update the model, an impulse response function can be used to estimate how a shock might affect inflation. When adding fuel price index to our data set, due to the index rising year on year. It becomes inadequate for VECM so a VAR is used which does not need a steady state equilibrium.

A shock of 8.9% in fuel prices produced a 0.55% percentage point increase in inflation which seems on the low end due to the drastic increase in fuel prices. Another war in the middle east in 1990 was similarly as short as this one is predicted to be and had a 19% increase before returning to pre war levels. This shock produced a 1.27% percentage point increase which is closer to the magnitude of what we are currently experiencing.

